Environmental Innovation and Societal Transitions 10 (2014) 4–19

Contents lists available at ScienceDirect

Environmental Innovation and Societal Transitions

journal homepage: www.elsevier.com/locate/eist

The concept of "lead markets" revisited: Contribution to environmental innovation theory

Rainer Quitzow[a,*], Rainer Walz[b,1], Jonathan Köhler[c,2], Klaus Rennings[d,3]

[a] Environmental Policy Research Centre (FFU), Department of Political and Social Sciences, Freie Universität Berlin, Ihnestr. 22, 14195 Berlin, Germany
[b] Competence Center Sustainability and Infrastructure Systems, Fraunhofer Institute for Systems and Innovation Research (ISI), Breslauer Str. 48, 76139 Karlsruhe, Germany
[c] Competence Center Innovation and Technology Management and Foresight, Fraunhofer Institute for Systems and Innovation Research (ISI), Breslauer Str. 48, 76139 Karlsruhe, Germany
[d] Department of Environmental and Resource Economics, Environmental Management, Zentrum für Europäische Wirtschaftsforschung, L7, 1, 68161 Mannheim, Germany

ARTICLE INFO

Article history:
Received 17 December 2012
Received in revised form 1 October 2013
Accepted 17 November 2013

Keywords:
Competitiveness
Emerging economies
Environmental innovation
Geography of innovation
Sustainability transitions

ABSTRACT

The economic potential of lead markets for environmental technologies has served as an important justification for policies to promote environmental innovations in Europe. The emergence of competition from emerging economies has revealed that a domestic lead market is no guarantee for long-term competitive success in an international context. This article reviews the academic literature on lead markets as well as its application in policy. It outlines the theoretical foundations of the concept and main insights obtained with it and relates these to its uses in policy. It then contrasts the lead market approach with the technological innovation systems framework and the multi-level perspective as these are both frequently employed in the transitions literature. The article addresses shortcomings of the current literature and proposes a number of avenues for further development of the concept.
© 2013 Elsevier B.V. All rights reserved.

* Corresponding author. Tel.: +49 30 838 56688; fax: +49 30 838 56685.
E-mail address: rainer.quitzow@fu-berlin.de (R. Quitzow).
[1] Tel.: +49 721 6809 236; fax: +49 721 6809 135.
[2] Tel.: +49 721 6809 377; fax: +49 721 6809 330.
[3] Tel.: +49 621 1235 207; fax: +49 621 1235 226.

2210-4224/$ – see front matter © 2013 Elsevier B.V. All rights reserved.
http://dx.doi.org/10.1016/j.eist.2013.11.002

1. Introduction

The rapid development and international diffusion of environmental innovations is key to sustainable development. The threat of irreversible climate change and increasing resource scarcity have helped to bring this issue to the attention of a growing community of policy makers and researchers. In the European Union, for instance, the concept of eco-innovation has become an important issue of policy attention, as manifested by the Eco-Innovation Action Plan (European Commission, 2011) or the various efforts of the European Commission to promote environmental innovation and related R&D (Quitzow, 2011). Outside the EU, emerging economies, including China and India, are developing environmental technology sectors of their own. In 2011, China led the world in terms of investments in renewable energy technologies (UNEP, 2012), and it currently even tops the Renewable Energy Attractiveness Index published by Ernst and Young (2012).

The increasing engagement of emerging economies with environmental technologies has gone hand in hand with growing competition in the field. Both industrialized and emerging countries are increasingly positioning themselves to capture leadership positions in these industries. The EU's Europe 2020 growth strategy places particular emphasis on maintaining the EU's leadership position in developing green technologies (European Commission, 2010). Similarly, in May 2012, the Chinese government passed a plan to promote seven strategic emerging industries, including environmental protection and energy-efficiency technologies, renewable energy, and so-called new-energy vehicles (Climate Group, 2011).

In this context, eco-innovation policy is no longer viewed merely as a subset of environmental policy, but it represents an increasingly strategic element of a new generation of industrial policy. A unique feature of this new industrial policy is that it does not suffice to promote the competitiveness of national industries or technology clusters. In the absence of a self-sustaining market for environmental technologies, government intervention cannot limit itself to supply-side interventions, such as R&D policy or other incentives for technology development. Equally important are demand-side measures that promote market development.

A key concept that has entered the policy discourse in this context is the notion of a lead market. Policy makers, particularly in the EU and in Germany, have eagerly employed the term to justify demand-side measures aimed at promoting renewable energies or other environmental technologies. The promotion of lead markets for environmental technologies, so the argument goes, can secure first mover advantages for national firms.[4] In this way, the lead market concept has made an important contribution to the development and diffusion of renewable energy technologies. By functioning as a justification for pioneering policies for promoting renewable energy in Germany, it has helped pave the way for international diffusion of technologies and rapid industrial development in the sector. At the same time, its alleged promise to secure a lasting competitive advantage for German companies has not always been fulfilled (Federal Ministry of Education and Research, 2011). The recent turmoil in the German solar industry is a case in point.

Against this background, this article seeks to revisit the lead market concept, placing it in the context of the broader literature on environmental innovation and technological change. The article begins with a brief history of the concept and an overview of its theoretical foundations and key policy applications. After this introduction, it discusses its contribution to the study of environmental innovation, contrasting it to the technological innovation system (TIS) framework and the multi-level perspective (MLP) from the transitions literature. Next, the article addresses shortcomings of the concept and proposes a number of avenues for its development and improvement. The paper ends by returning to the politically relevant discussion on environmental innovation and economic competitiveness and proposes important questions for further research.

[4] A detailed economic rationale for promoting lead markets is provided in the European Commission Staff Working Paper "Explanatory Paper on the European Lead Market Approach: Methodology and Rationale" (European Commission, 2007a).

2. The development of the lead market concept

2.1. The emergence of the concept

The lead market concept first emerged in the business management literature. Lead markets were identified as important learning and testing grounds for innovations developed by multinationals. In his article on *The Changing Patterns of International Competition*, Michael Porter (1986) identifies that the particular demand conditions of a country may make it particularly suitable for launching an innovation in that market. Bartlett and Ghoshal (1990) further developed this insight and linked it to the lead market terminology. According to them, lead markets represent "the largest, most sophisticated and most competitive market in which the nature of impending global changes is mirrored" (Bartlett and Ghoshal, 1990, p. 242). Relating it to strategic options for the management of innovation in multinational firms, they argue that success in these lead markets will have a determining influence on the competitive position of firms in the related sector. Bartlett and Ghoshal's use of the concept largely reflected an empirical observation without any explicit links to theories of innovation and technological change. They made no attempt to systematically define the concept, giving rise to a number of descriptive uses of the term. Scholars at the time used the term to denote the country where a technology was invented (Yip, 1992), where it first experienced success in the market (Kotabe and Helsen, 1998) or where a multinational has placed global responsibility for a certain product group (Raffée and Kreutzer, 1989).

The first attempts to elaborate a more nuanced definition of the lead market concept were made by Meyer-Krahmer (1997) and Meyer-Krahmer and Reger (1999). Their starting point was a large empirical research project on the localization of R&D activities of multinational companies. Their empirical findings corroborate the importance of linking R&D activities to lead markets. Interpreting their results, they derive a first characterization of lead markets. In their own words, they describe a lead market as a market characterized by one or more of the following characteristics (Meyer-Krahmer and Reger, 1999, p. 771):

1. "a demand situation characterized by high income elasticity and low price elasticity or a high per capita income;
2. a demand with high quality requirements, great readiness to adopt innovations, curiosity concerning innovations and a high acceptance of technology;
3. good framework conditions for rapid learning processes by suppliers;
4. authorization standards that are 'setting standards' for permit authorization in other countries, (e.g. pharmaceuticals in the US);
5. a functioning system of exploratory marketing ('lead user' principles);
6. specific, problem-driven pressure to innovate;
7. open, innovation-oriented regulation and framework conditions."

The work of Meyer-Krahmer and Reger did not, however, aim at developing a definition of lead markets from a specific theoretical foundation. It represented a heterodox interpretation of empirical findings alluding to concepts such as the evolutionary theory of trade (Dosi and Soete, 1988; Fagerberg, 1996), Porter's (1990) competitive advantages of nations, the lead user principle of Von Hippel (1986, 1994), or the learning economy hypothesis of Lundvall and Johnson (1994). No systematic effort was made to integrate them within a coherent conceptual framework.

2.2. Development of an analytical framework and its connection to evolutionary economics

The theoretical elaboration of the emergent lead market concept into a more clearly defined framework for analysis was undertaken by Beise (2001, 2004, 2006). Beise defines a lead market as a geographical market where local demand preferences and local environmental conditions induce the emergence of an innovation design that diffuses globally. A lead market defined in this way does not refer to the country where a product or technology was invented, but rather where it had significant market success before diffusing globally. So far, Beise's definition is compatible with the definition

offered by Meyer-Krahmer. In the process of analytical refinement, however, Beise introduces a number of assumptions and conceptual clarifications, which narrow the scope of the concept vis-à-vis its predecessors. Firstly, Beise explicitly excludes supply-side aspects from his discussion. While he acknowledges that aspects, like technological capacities (Dosi and Soete, 1988; Fagerberg, 1996) may help explain the global success of innovations, he argues that in most cases such factors do not play a critical role (Beise, 2006, p. 121). Meyer-Krahmer and Reger (1999) on the other hand allude to the performance of existing supplier networks as a factor in explaining lead market development. Secondly, Beise explicitly rejects Meyer-Krahmer's claim that lead markets are determined by an affinity to try out new innovations and accept technological advances (point 2 in the list above) (Beise, 2006, p. 115). Instead he stresses that the global diffusion of an innovation is typically preceded by competition between several local innovation designs. One example that he provides is the parallel emergence of the minitel in France and the internet in the US (Beise, 2004, p. 999). What sets a lead market apart from other countries is not that it is the only market where an innovation emerges, but that it becomes the source of a global "dominant design."

The dominant design concept, as defined by James Utterback (1994), represents the central theoretical underpinning of Beise's lead market framework. Dominant designs represent innovation designs that serve as a standard, based on which incremental innovations are pursued in an industry. Until a dominant design emerges, a diverse set of technological options will develop in parallel and radical innovations are more likely. Once a dominant design is in place, competition begins to shift more strongly towards cost and scale elements and incremental improvements of product performance (Abernathy and Utterback, 1978; Anderson and Tushman, 1990; Utterback, 1994).

As already indicated, a lead market represents a market where the diffusion of a dominant design first takes off. It points to the fact that technological change is not only determined by sectoral dynamics but also by the segmentation of innovative activities in different geographic entities. This geographic interpretation of the dominant design concept represents the point of departure for Beise's lead market approach. On that basis, Beise elaborates five mutually-reinforcing success factors to help explain the emergence of a global dominant design from a particular country (or in some cases region, e.g. California). They are referred to as "lead market advantages," as they help explain the competitive success of an innovation from a lead market vis-à-vis similar innovations in other "lag markets." Drawing on different concepts from evolutionary economics and innovation research, each lead market advantage synthesizes a different set of factors influencing the process of international diffusion and links these factors to underlying country-level attributes. They are labelled as the demand, price, market structure, transfer and export advantage. In the remainder of this section, each lead market advantage and its most important theoretical underpinning are briefly presented.

The demand advantage refers to the fact that environmental conditions in the lead market may induce an early demand for a certain product or service, conferring an early mover advantage onto firms in the lead market. This early demand hypothesis corresponds to the main explanation for the lead market phenomenon offered by Bartlett and Ghoshal (1990). They state, "local innovation in such [lead] markets becomes useful elsewhere as the environmental characteristics that stimulated such innovations diffuse to other locations" (p. 243). Moreover, it builds on and extends Vernon's product-life-cycle theory (Vernon, 1966). Among other things, Vernon proposed that higher incomes in the US enabled product innovations, which later diffused to other countries after average incomes reached similar levels. Beise extends this to all types of social, environmental and economic trends, developing at different paces in different countries. His reasoning displays important parallels to the concept of "anticipatory demand" in Michael Porter's work on the competitive advantage of nations (Porter, 1990).

While the demand advantage is a core feature of the lead market concept, Beise stresses that alone it is not sufficient for explaining a lead market. In fact, he places even greater emphasis on the price advantage (Beise, 2004, p. 1003), referring to two key dimensions: Firstly, it addresses the fact that the market, which achieves early cost reductions, has a better chance of becoming the global dominant design. The key driver of this is the ability to capture economies of scale – either via early expansion of local demand or via a large domestic market. Secondly, the price advantage relates to trends in relative input costs, which may confer an advantage to a country that anticipates a particular cost

trend. According to this logic, the innovation design in the frontrunner country becomes competitive in other countries once the relevant cost trends internationalize.

The next lead market factor, the market structure advantage, is essentially borrowed from Porter's concept of "rivalry" (Porter, 1990) and refers to the level of competition within a national market. The reasoning is that increased competition will enhance innovation and make firms more responsive to customer needs. Resulting products are expected to offer a better cost-benefit relationship than products from less competitive markets. In addition, competition tends to push down the overall price of a product thus reinforcing the price advantage.

Finally, the last two lead market factors relate to the ability of a country to influence consumers in foreign countries (transfer advantage) and the ability or tendency to respond to customer needs in these foreign countries (export advantage). The former mainly builds on the "demonstration effect" commonly referred to in diffusion research (Mansfield, 1968). Beise proposes that the strength of the demonstration effect may vary depending on the country of origin of the innovation (due to aspects such as reputation or overall importance in the global economy). The latter relates primarily to the export orientation of a particular country and the tendency of firms to consider preferences in important foreign markets. While this factor is not clearly linked to a specific concept in the innovation literature, Beise supports it with a number of empirical examples.

2.3. Lead markets for environmental innovations

Building on the lead market approach described above, Jacob et al. (2005) have undertaken additional research to identify the particularities of lead markets for environmental innovations. The research integrated Beise's lead market approach with findings from the existing literature on the relationship between environmental regulation and innovation as well as policy diffusion. The research confirmed that lead markets, like markets for environmental innovations more generally, are typically induced by well-designed environmental regulations, which stimulate a related technological innovation. The subsequent international diffusion of these innovations may follow two patterns. If the innovation offers additional economic benefits, it is likely to diffuse via market mechanisms and the original lead market factors apply. If this is not the case, the international diffusion of the technological innovation is typically preceded by the international diffusion of the regulation (i.e. the adoption of the regulation by follower countries), which induced the original innovation in the lead market (Beise and Rennings, 2005; Jänicke and Jacob, 2004). In the latter case, lead markets for environmental innovations are likely to form when environmental regulations introduced in a pioneer country anticipate (or help to set) international trends in environmental policy.

To capture this phenomenon, Rennings and Smidt (2010) speak of a "regulatory advantage." Complementing the remaining lead market advantages, the regulatory advantage highlights that environmental regulations; which tackle environmental challenges of increasing global relevance, may induce the development and international diffusion of corresponding environmental innovations via the mechanisms outlined above. The countries who are sponsors of such anticipatory regulations may thus develop into lead markets for these environmental innovations.

3. The lead market approach in policy

Drawing on their respective interpretations of the lead market concept, a number of the authors cited above have also developed policy conclusions. In particular, they have discussed the potential competitive advantage conferred on firms or industries in the lead market. Meyer-Krahmer (2004) highlights the fact that a country or region cannot simply rely on research excellence and technological leadership to promote a long-term competitive advantage in a given technology field. Only in combination with the points cited in Section 2.1 is it likely that a country or region can position itself as a location for core technological competences (see also Walz and Schleich, 2009; Walz, 2006). By the same token, however, he points towards the conclusion that the existence of a lead market alone does not automatically secure economic advantages for the country. Beise et al. (2002) argue that demand-side aspects should play a prominent role in development of an effective innovation

policy. They emphasize the importance of competition within the domestic market and support the formulation of innovation policies based on anticipated societal trends. In the field of environmental innovation, Jänicke and Jacob (2004) highlight the possible competitive advantages of policy-induced lead markets. Citing the Porter Hypothesis (Porter and Van Der Linde, 1995), they state that pioneering environmental regulations may provide industries in the lead market with early mover advantages via "learning effects or patent protection for the innovation" (Jänicke and Jacob, 2004).

These policy-oriented discussions quickly attracted the attention of policy makers. By 2006, the lead market concept had been adopted as the guiding vision for the German government's "Hightech Strategy" led by the Ministry of Education and Research (BMBF) where Meyer-Krahmer had assumed the position of vice-minister (*Staatssekretär*). Largely following his lead market approach, the strategy promotes four lead markets based on important societal needs, i.e. health, security, mobility and climate, resource preservation and energy, in conjunction with 17 so-called "key technology" fields. Moreover, it highlights the importance of an integrated innovation policy, spanning the full spectrum of policy fields and combining demand-side measures, such as standardization and public procurement, with more traditional supply-side policies (Federal Ministry of Education and Research, 2006).

Parallel to these developments in Germany, the concept took hold in the European Commission. Its Competitiveness Report 2006 highlights a series of competitive advantages associated with a lead market: "firms may profit from increasing exports, foreign direct investment may be attracted, further research may be stimulated, or productivity gains may be realized earlier than in other countries." (European Commission, 2006b, p. 111). A number of reports and strategy papers on innovation and industrial policy call for measures aimed at the development of European lead markets (European Commission, 2006a, 2006c, 2007b), culminating in the launch of the "lead market initiative" (LMI) in 2007. In the accompanying explanatory paper, the Commission's adopts a slightly expanded interpretation of Beise's definition:

> "A lead market is the market of a product or service in a given geographical area, where the diffusion process of an internationally successful innovation (technological or non-technological) first took off and is sustained and expanded through a wide range of different services" (European Commission, 2007a, p. 12)

In order to stimulate such lead markets, the LMI proposes an "integrative co-ordination of mutually inter-dependent policy measures," which act on "the demand-side factors" of innovation.

The most optimistic interpretation of the lead market concept was adopted by Germany's environment ministry in its "ecological industrial policy" (Federal Ministry of the Environment, 2006). Building on the concepts of Jänicke and Jacob (2004), the Social Democrat-led ministry developed a vision of environmentally-friendly lead markets as the basis of future economic development and employment generation. The ministry's report to parliament on its "masterplan" for the promotion of environmental technologies states:

> "Frequently, environmental regulations, which were first introduced in a particular country, are later adopted by other countries and have thus provided the firms of the pioneer country with export advantages. Environmental regulation can thus become an instrument, which lets a "lead market" for innovations emerge." (Government of Germany, 2008b, p. 3)

In contrast to the BMBF and European Commission, this use of the lead market concept served not merely as a complement to a broader innovation strategy, but also helped to justify an ambitious set of demand-side policies to support the development and diffusion of renewable energies and a range of other environmental technologies (Government of Germany, 2008a).

4. The lead market concept and the study of environmental innovation

As outlined in the preceding sections, the lead market approach has played a major role in highlighting the importance of demand-side factors, both in innovation research as well as in policy. This section focuses particular attention on the contribution of the concept to the study of environmental innovation, contrasting it to approaches from an innovation systems (IS) perspective, in particular the

technological innovation systems (TIS) framework, and the multi-level perspective on socio-technical transitions (MLP). It argues that it adds a unique perspective to the field by highlighting geographic aspects of environmental innovation, while relating them to aspects of international competition.

As noted above, the approach seeks to explain *why* an (environmental) innovation introduced in one specific country spreads to other countries and replaces other technologies there. It focuses on the distinct patterns of innovation across countries (and to lesser extent regions) and employs this to explain the international diffusion of certain innovations within a competitive global economy. By doing so, it provides a framework for explaining why environmental innovations emerging in one country become globally successful, while others remain confined to a particular geographic niche. As such, it also places particular emphasis on the competitive dynamics between different emerging technologies rather than between new and old technologies.

Within the broader systems of innovation research agenda, the geographic dimension of innovation processes has been captured in the original National Innovation System (NIS) concept (Lundvall, 1992; Nelson, 1993). However, NIS research does not focus on how a particular NIS has influenced the competitive dynamics within a particular sector.[5] Rather it has analyzed the configuration of different NIS and pointed out their relative strengths and weaknesses in shaping the competitive advantage and general patterns of specialization of market economies. Consequently, it has also not considered the roles of different NIS in the development and diffusion of specific environmental technologies. Conversely the sectoral systems of innovation (SSI) approach focuses mainly on actors involved in innovation and their inter-linkages within a given sector, but it does not emphasize how these are shaped by country-specific dynamics. Rather it has focused on comparing similarities and differences across sectors, both in terms of the configuration of the SSI and the related dynamics of innovation (Malerba, 2006, 2004). Moreover, with a few exceptions, it has not been applied to analyze environmental innovation (Köhler et al., 2012; Rogge and Hoffmann, 2010).

Within the IS research agenda, the TIS is the main framework for analyzing environmental innovations. It has been utilized to study the evolution of emerging technology-specific innovation systems, while also considering country-specific dynamics. However, as pointed out in Coenen et al. (2012), empirical applications of the TIS have frequently focused on single countries. The existing multi-country studies have been comparative in nature, aiming to describe the specificities of individual countries. Although it has been pointed out that, in principle, a TIS spans geographic boundaries (Carlsson and Stankiewicz, 1991; Hekkert et al., 2007), these studies have not provided any systematic treatment of cross-country interactions or other international influences on the development of a country-specific TIS. Rather most studies treat each country-specific TIS as an autonomous system operating within the same "global opportunity set" (Carlsson et al., 2002). The lead market approach offers an important entry-point for addressing this blind spot in the TIS literature.

Finally, the MLP may be considered the broadest in scope of the three perspectives presented here. Rather than focusing on competition between individual innovation designs or between specific technological artefacts, the focus lies on understanding system innovations or sustainability transitions, representing shifts in the broader socio-technical regime (Elzen et al., 2004; Rip and Kemp, 1998; Geels, 2002; Grin et al., 2010). The critical idea is that radical innovations often first manifest themselves in niches, which under favourable circumstances may grow and challenge the accepted way of doing things – the socio-technical regime. Using a relatively long-term, historical perspective, the MLP has been applied to analyze a wide range of socio-technical transitions. In doing so, most applications of the MLP have – similar to TIS studies – focused on the national level. According to Geels, this may be explained by the fact that "cultures, infrastructures, regulations and institutions are mostly (but not always) national phenomena" (Geels, 2011, pp. 13–14). Simultaneously, however, the core MLP concepts (i.e. niche, regime and landscape) are explicitly not geographical in their nature, but related to the maturity of the socio-technological system (Coenen et al., 2012). The role that the geographic locus of developments plays in the transition process is not addressed in detail. Though individual studies (see for example Geels (2002) study of the transition from sailing ships

[5] A conceptual contribution has been made, which links a NIS perspective to concepts that have emerged from research on global value chains (Pietrobelli and Rabellotti, 2011).

Table 1
Key differences between the lead market approach, the multi-level perspective and approaches from the literature on innovation systems.

Framework	Aim	Role of environmental innovation	Role of geography	Role of competition
Lead market approach	To explain the emergence and diffusion of global dominant designs	Highlights the special case of environmental innovations and the role of regulation in their diffusion	Focused on the role of country-specific factors in shaping the competition between emerging technologies	Focused on the inter-relationship between country- and technology-specific competition with a focus on emerging technologies
Multi-level perspective	To analyze long-term technological change, i.e. shifts in technological regimes	Provides a framework for understanding transitions to more sustainable socio-technical regimes	Not explicitly captured theoretically; Empirical studies mainly focus on single countries	Focus on the competition between different technological regimes
Technological innovation systems	To analyze the dynamic development of emerging technologies	Suitable for the analysis of emergent environmental innovations	Not captured theoretically; Empirical studies focused on single countries or comparisons	Not an explicit focus
National innovation systems	To analyze and compare country-specific institutions, actors and their interactions and how they shape innovation in a country	Not a focus	Single country studies and multi-country comparisons	Country studies point out strengths and weaknesses of country-specific NIS and help explain broad patterns of specialization
Sectoral systems of innovation	Analyze and compare the dynamics of innovation at sectoral level	Focus on established sectors	Country-specific dimensions considered but not the focus	Country-specific specialization considered but not the focus

to steamships) do acknowledge parallel developments in different countries, they lack a systematic treatment of how differing national-level dynamics interact and influence processes of innovation and technological change at the international level. Moreover, they pay little attention to the role that geographic entities play in providing (protected) spaces for the development of competing innovations. Table 1 provides a summary of the most important differences across the various frameworks, highlighting their respective treatment of environmental innovation, geography and competition.

The focus in the lead market literature on how country-level characteristics may influence international processes of technology diffusion thus also provides an important entry point for further enriching the MLP. Walz and Köhler (2014) propose that the development of a lead market can be interpreted as a growth path of a technological niche, shaped in part by the perspective of exporting the technology to other countries. From an aggregated, country-level view, they suggest that the demand, transfer and export advantage can be operationalized with indicators, which they assign to the landscape level in the MLP, such as awareness of sustainability, general technological competences ascribed to the country, and openness of the home industry to foreign preferences. When referring to a specific technology, on the other hand, the transfer and export advantage would be strongly influenced by the performance of the niche technologies and the country's position within the given sector. Thus, these advantages would also be shaped at the niche level. Similarly, the price and regulatory advantages depend very much on sector and technology specific developments. They represent key factors in the interplay between niche and regime in environmental technologies. International trade represents

an inter-linkage between niches in different countries, enabling knowledge spillovers and learning by exporting or importing. Finally, international policy learning and diffusion can be interpreted as a link between the interplay of niche and regime in different countries.

5. Critical review of the lead market model

The ability of the lead market concept to capture country-specific factors in explaining the global competitive success of innovations has been an important reason for the high level of political traction it has received. It also suffers from important conceptual weaknesses, however, a number of which have come to the fore as a result of the German policy experiments it has helped in justifying. The following section explores several of these shortcomings, drawing on a number of examples from the German solar energy sector.

5.1. Applicability of the dominant design concept in the context of the lead market framework

Firstly, as outlined above, the analytical framework promoted by Beise relies heavily on the concept of a dominant design. However, Beise does not address the question to what type of technological innovations the dominant design concept and hence the lead market concept is applicable. Although Beise does not claim universal applicability of the lead market concept, a more nuanced analysis of the question of applicability would add significantly to its analytical value. Utterback and Abernathy (1975), for instance, restrict their conclusions regarding the emergence of dominant designs to complex assembled products. Others have suggested that they are limited to mass markets with relatively homogeneous consumer demand (Teece, 1986; Windrum and Birchenhall, 1998) or complex, systemic technologies (Nelson, 1995). Moreover, it has been acknowledged that dominant designs may attain different degrees of market penetration, and it is a subject of dispute what level of market penetration is required for the term to apply. Finally, Murmann and Frenken (2006) point out that dominant designs can occur at different levels of a technological hierarchy and may be replaced at differing rates, depending on the duration of the technological cycle.

A number of possible implications may follow from these issues. The examples that Beise provides in his work are primarily innovations that can be linked to a specified demand by the end consumer, such as the fax machine or cellular phones. These fit neatly with the lead market approach, which builds heavily on the demand advantage for explaining the shape of a particular innovation design. However, innovation and dominant designs can also emerge in parts of an established technology or product class without significantly changing the use value of the product in question, apart from further enhancing an existing feature (such as increasing robustness or lowering the price). Arguably, these types of innovations can be considered incremental in the context of the lead market model. However, in the particular technological niche, they may well represent a more disruptive innovation and the replacement or creation of a dominant design, which may have significant implications for related industry players. An example of this is the competition between thin-film and silicon-based solar modules. While the functional use of the solar module remains essentially the same, the underlying technology may be very different. As a result competition is not based on an innovation design shaped by the environmental conditions in any particular country but rather by innovation in the technology and the production process, i.e. by issues related to the supply-side.

On the other side of the spectrum, changes from one dominant design to another at the highest level of a technological hierarchy may imply such far reaching changes that whole systems are reconfigured, implying far reaching social and institutional changes taking place over prolonged periods of time. In the literature on sustainability transitions, these have been called regime shifts or system innovations (Elzen et al., 2004; Geels, 2002; Grin et al., 2010; Rip and Kemp, 1998). It may be fairly self-evident that the narrowly defined lead market concept developed by Beise with its focus on price mechanisms and demand preferences does not extend to this broad type of system innovation. Nevertheless, elaborating more specifically which type of innovations may be explained by Beise's lead market advantages and which innovations are beyond its scope represents an important avenue for enhancing its analytical

value. It would also help position it more clearly in the broader literature on environmental innovation and transitions to sustainability.

5.2. Regulatory advantage and political issues

The regulatory advantage in the lead market approach highlights the importance of policy measures in the development of lead markets for environmental innovations. While policy measures and their diffusion play a key role in this context, it is argued here that the current definition of the regulatory advantage does not do justice to the complexity of the process of environmental innovation. In the following, two important aspects, which are not sufficiently reflected in the concept, are highlighted.

Firstly, the concept of regulatory advantage stresses the importance of individual policies (such as feed in tariffs or classic environmental regulations) in explaining the development of lead markets for environmental innovations (Beise and Rennings, 2005; Jänicke and Jacob, 2004; Rennings and Smidt, 2010). It does not emphasize the fact that the emergence of environmental innovations typically involves some form of supply-side intervention (Jänicke and Lindemann, 2010). A classic regulatory intervention (such as tightening of environmental standards) alone is rarely sufficient for explaining the emergence of a lead market. Consequently, the innovation design which emerges, is shaped by a broader policy mix as well as other demand conditions. Moreover, the regulatory advantage might extend beyond the policy-level to industry-specific product standards or testing procedures, which may offer a lead market advantage if adopted internationally.

Secondly, as indicated above, studies have found that lead markets frequently take shape by virtue of anticipatory policies which are later adopted by other countries (Jänicke and Jacob, 2004). However, what these studies have failed to consider is to what extent a dominant design has in fact emerged at the time of policy diffusion. Policy-driven diffusion processes may precede the emergence of a dominant design, in which case it merely enables competition between different innovation designs. In other words, the market, which pioneers the environmental policy, may or may not represent a lead market, as defined by Beise.

Once again the solar energy sector offers a good example. While Germany is a pioneer market in terms of the adoption of solar energy, it is most likely not going to emerge as the market driving the further development of the innovation design. When variations of the German feed-in tariff were imitated by countries like Spain and Italy, no dominant design had emerged. There was still competition, most importantly between thin-film and silicon-based solar panels. While silicon-based modules appear to be emerging as the dominant design, it is unlikely that future incremental innovations will evolve according to conditions in the German market (i.e. characterized by cold weather and little sun). It is more likely to evolve based on climatic conditions in countries offering more significant solar potential. The key point here is that policy dynamics may in fact distort early diffusion patterns, which may be corrected once the technology becomes competitive without policy support.

5.3. The role of supply-side dynamics

Finally, a key question regarding the analytical value of Beise's lead market concept relates to the role that supply-side aspects play in shaping lead markets. While Beise is explicit about the fact that his framework does not take into consideration supply-side dynamics (Beise, 2006, p. 121) this paper argues that these aspects are too important to be ignored. They bear an important relationship to what Coenen et al. (2012) address in their "geographical critique" of transition studies. The authors point to the fact that both the territorial embeddedness ("local nodes") and the global dimension of innovation systems ("global networks") require a more explicit acknowledgement in the transitions literature. Similarly, this paper suggests that the local and global dimensions of innovation and production systems are highly relevant for the lead market framework as well as its implications for national competitiveness.

Arguably, supply-side aspects are already partially integrated in the lead market advantages, making it even more pertinent to explicitly acknowledge their role in the framework. For instance, the export advantage refers to the export orientation of local industries. Although this may be a more general characteristic applicable to entire countries, it will always display variance across different

industries within a country. Relevant to lead market development, however, is primarily the export-orientation of the specific industry where a particular innovation emerges. It might even be argued that the export-orientation (read: export-success) of an industry is, in fact, a proxy for the existence of a competitive industrial cluster.

The importance of supply-side dynamics in explaining the international success of innovations is also well established in innovation research. In the 1980s, the evolutionary theory of trade was developed by scholars such as Dosi, Soete, or Fagerberg. Since then it has become increasingly accepted that international trade performance depends on technological capabilities (for an overview see Dosi and Soete, 1988; Fagerberg, 1996). This result is also important for the lead market debate. Despite the problems and caveats associated with measuring technological capabilities, indicators on R&D expenditures and patents are among the most widely used, and their relevance is supported by recent empirical research (e.g. Sanyal, 2004; Andersson and Ejermo, 2008; Madsen, 2008). It is also widely accepted by scholars of evolutionary economics that innovation and economic success depend on how a specific technology is embedded into other relevant industry clusters. Learning effects, expectations of the users of the technology and knowledge spillovers are more easily realized if the flow of this – very often tacit – knowledge is facilitated by proximity and a common knowledge of language and institutions (Archibugi and Pietrobelli, 2003; Dosi et al., 1990; Fagerberg, 1995)

Regarding the specific implications of such supply-side aspects for the lead market concept, this paper highlights three (partially inter-related) dimensions. The first relates to differences in how global production networks are arranged in different types of industries or technological fields. In other words, it relates to the geographical distribution of the global TIS. The second relates to the technological capabilities and supplier networks that are embedded in a specific territorial entity, which, as will be argued, has implications for the lead market potential of a country or region. Finally, an additional factor in the case of environmental innovations relates to the interaction between supply-side dynamics and regulatory design.

In reference to the first aspect, Beise's lead market concept implicitly assumes the industry structure proposed in Vernon's original product-life cycle theory (Vernon, 1966, 1979). The underlying assumption is that in early phases of the product life cycle, R&D and production are located in the market where an early demand for the respective product emerges. This assumption is essential in applying the lead market advantages as suggested by Beise. For instance, the price advantage depends on the assumption that supply and demand are located in the same country, thus conferring a scale advantage on larger countries. However, in sectors where markets are contested by firms that operate at a global scale (at least in terms of sales), the price advantage may be dominated by economies of scale or scope that are independent from the demand dynamics in any single country. In fact, the latter situation is becoming the norm in an increasing number of cases.

The challenge of capturing the complexities of the international system of production is not unique to the lead market concept. Porter's work on the competitive advantage of nations has also been criticized for its failure to consider the role of multinationals, which may have production sites spread out across a number of different markets (Rugman, 1991). A useful starting point for extending the lead market concept to more internationalized markets may, therefore, be to revisit (i.e. further refine and systematically test) the approach offered by Meyer-Krahmer and Reger (1999) whose findings build on the investment behaviour of multinationals. In addition, further research is needed to better understand and link specific sectoral characteristics to the resulting patterns of internationalization in production. Empirical results indicate that the drivers of value chain internationalization vary across industries, resulting in different patterns of internationalization. In more knowledge-intensive sectors, for instance, unit labour costs play a relatively smaller role in determining competitiveness (Amable and Verspagen, 1995; Wakelin, 1998). Instead intensive user-producer relationships and the locus of tacit knowledge become more important (Archibugi and Michie, 1998; Archibugi and Pietrobelli, 2003; Dosi et al., 1990; Fagerberg, 1995). Finally, the literature on global value chains and their differing compositions and modes of governance offers a number of lessons that might be leveraged for developing a more nuanced understanding of the lead market advantages and their applicability in different industrial sectors (Pietrobelli and Rabellotti, 2011).

Tiwari and Herstatt (2012a) have made a first attempt at considering the role of supply-side aspects for lead market development in a number of empirical studies on low-cost innovations in India. They

analyze electronic voting machines and small cars, and conclude that "in both instances the lead market in question enjoys significant technological capabilities, which have played a key role in the emergence of India as a lead market" (Tiwari and Herstatt, 2012a). Extending the analysis to water purifiers, solar powered ATMs and battery powered refrigerators, they confirm the crucial role of supply side aspects such as availability of skilled labour, R&D capacities and access to knowledge or technologies (Tiwari and Herstatt, 2012b). They summarize these factors as "technological advantage."

While offering a first empirical basis for integrating supply-side aspects in the lead market concept, it will require additional research to provide a more systematic assessment of when and how supply-side dynamics may influence both the development and retention of lead markets. Continued research on low-cost innovations in China, India and other large emerging economies offers one set of promising empirical cases in this context. Emerging findings, from the business management literature are revealing that the composition of supplier networks and the dynamics of interaction among the related firms play a key role in delivering so-called frugal innovations in India (Bruck, 2011; Sehgal et al., 2010). Similarly, the dense network of highly specialized electronics suppliers in China's Pearl River Delta has provided the basis for innovation and export success in generic and very low cost cellular phones (Kao and Lee, 2010; Zhu and Shi, 2010).

Finally, in the case of environmental innovations, feedbacks between policy interventions and existing supplier groups further enhance the particular role that supply-side dynamics play in shaping country-specific technology trajectories (or innovation designs). The importance of policy interventions for promoting environmental innovations implies that market developments are shaped not only by market dynamics, but by the design and choice of policy interventions. This in turn is heavily influenced by the political process, in which domestic technology suppliers and economic considerations related to the domestic industry may play a key role (Jänicke, 2012). Policy measures that offer new opportunities to domestic industries are more likely to be approved than those that are likely to offer significant opportunities to foreign companies or that will compete with products from existing industry players.

The strategies for promoting electric vehicles in Germany and China represent a practical example of this phenomenon. While the German government pursues the declared goal of developing a domestic lead market for electric vehicles, the related policy mix currently remains skewed towards supply-side measures, such as R&D and skill development, targeted at Germany's existing car manufacturers (Acatech, 2010). The German National Platform E-Mobility aims explicitly at becoming a market with high lead market potentials, however, it has postponed broad market support instruments to a later stage (Nationale Plattform Elektromobilität, 2011). China, on the other hand, whose strength lies primarily in battery rather than car manufacturing, has been more willing to invest in demand-side policies, hoping to secure a leadership position in a future electric vehicle industry (Zheng et al., 2012). As these examples suggest, policies in support of lead market development are strongly influenced by the domestic industrial base and related technology capabilities, further strengthening the case for the integration of supply-side aspects in the lead market approach.

6. Conclusion

In the preceding discussion it was argued that the lead market framework proposed by Beise is the most elaborated in terms of its conceptual underpinnings. After more careful investigation, it became clear that it is also somewhat limited in terms of its applicability. Most importantly, it implies that the successful innovation design displays the characteristics of a dominant design. In addition, the lead market advantages imply a relatively low level of internationalization in the initial stages of market formation. In the light of these (fairly general) conceptual qualifications, further research might help define a more specific and measurable set of criteria for assessing the applicability of the lead market concept to a given technology field. Defining more clearly the types of innovations to which the lead market concept can be applied or what the difference in application might be for different types of industries would add significantly to its analytical value.

An important question in the current context relates to the development of lead markets for policy-driven environmental innovations. The acknowledgement that policies and their diffusion play a crucial role in driving these markets represents a useful starting point. However, further research is

needed to explore the dynamics of diffusion processes in these policy-driven markets in more detail. Recent developments in the renewable energy sector have shown that policy-induced lead markets may be short-lived. As these markets are attaining competitiveness vis-à-vis incumbent technologies, China in particular is increasingly transitioning from a follower position to a major driver of these markets. While its early engagement was characterized by supply-side strategies to meet Western demand, today its markets for wind and solar energy are among the largest in the world. This raises the question regarding the stability of a lead market in the context of a policy-driven market. Will the market adjust to the "natural" lead market once the policy-driven phase of market development ends? How does the specific choice of policy influence these developments? To answer these questions, research into later stages of lead market development and diffusion is needed.

It has also been argued that lead market development is strongly influenced by supply-side factors. To account for this, a logical step would be the integration of supply-side aspects into the set of lead market advantages. This in turn would imply the elaboration of a lead market concept, which extends beyond innovations that develop into dominant designs. Such an approach might instead focus on the role of the lead market within a global innovation system and – possibly building on related concepts in the TIS literature – the functions it provides during different phases of system development and internationalization. Moreover, it might differentiate between different types of lead markets and their dynamics. Distinctions might be based, for instance, on the degree of geographical segmentation, the rate of internationalization of the supply chain or the pre-existence of closely related global value chains.

By both broadening and nuancing the lead market concept in this way, it would also offer a stronger framework for answering questions on the links between lead markets – in particular the policy-driven type – and competitive advantage. Identifying and explaining differing patterns of lead market development and diffusion would help gain a better understanding of the dynamic interaction between supply-side factors and other lead market advantages and how this influences competitive advantage. Rather than addressing the question of *whether* the presence of a lead market confers competitive advantage onto local firms, such an approach should help answer a number of more specific questions on lead markets and competitiveness. For instance, under what circumstances does lead market development coincide with the development of a lead supplier role? How does the success of local industry evolve over time and how does it differ across segments of the supply chain? How do incumbent industries influence lead market development? And from a policy perspective, what are the options for promoting competitiveness of local industry in combination with lead market development? Are their trade-offs between lead market and lead supplier policies?

Answering this set of more nuanced questions will fulfil an important political function. As described above, measures to promote markets for environmental innovations have been justified by policy makers in Germany and elsewhere by citing the promise of lead market development for building competitive advantage. Given the current shifts in the renewable energy industry, there is a danger that these overly optimistic assumptions may be replaced by equally simplistic assumptions about the potential economic costs of pioneering eco-innovation policy. An analytical approach with the aim of better explaining the dynamic inter-linkages between policy, lead markets and local industrial success will fill an important knowledge gap both in terms of this ongoing political debate and in terms of enhancing the design of eco-innovation policies.

Acknowledgements

This paper was developed within the context of a research project funded by the Germany Ministry of Education and Research entitled "Lead Market Strategies: First Mover, Early Follower and Late Follower" (www.zew.de/leadmarkets).

References

Abernathy, W., Utterback, J., 1978. Patterns of industrial innovation. Technology Review 50, 41–47.
Acatech, 2010. Wie Deutschland zum Leitanbieter für Elektromobilität werden kann, vol. 6.
Amable, B., Verspagen, B., 1995. The role of technology in market shares dynamics. Applied Economics 27, 197–204.

Anderson, P., Tushman, M.L., 1990. Technological discontinuities and dominant designs: a cyclical model of technological change. Administrative Science Quarterly 35, 604–633.
Andersson, M., Ejermo, O., 2008. Technology specialization and the magnitude and quality of exports. Economics of Innovation and New Technology 17 (4), 355–375.
Archibugi, D., Michie, J., 1998. Technical change, growth and trade: new departures in institutional economics. Journal of Economic Surveys 12 (3), 1–20.
Archibugi, D., Pietrobelli, C., 2003. The globalisation of technology and its implications for developing countries. Technological Forecasting and Social Change 70 (9), 861–883.
Bartlett, C., Ghoshal, S., 1990. Managing innovation in the transnational corporation. In: Bartlett, C., Doz, Y., Hedlund, G. (Eds.), Managing the Global Firm. Routledge, London, pp. 215–255.
Beise, M., 2001. Lead Markets: Country-Specific Success Factors of the Global Diffusion of Innovations. Physika-Verlag, Heidelberg.
Beise, M., 2004. Lead markets: country-specific drivers of the global diffusion of innovations. Research Policy 33 (6/7), 997–1018.
Beise, M., 2006. Die Lead-Markt Strategie: Das Geheimnis weltweit erfolgreicher Innovationen. Springer, Heidelberg.
Beise, M., Rennings, K., 2005. Lead markets and regulation: a framework for analyzing the international diffusion of environmental innovations. Ecological Economics 52 (1), 5–17.
Beise, M., Cleff, T., Heneric, O., Rammer, C., 2002. Lead Markt Deutschland: Zur Position Deutschlands als führender Absatzmarkt für Innovationen. Zentrum für Europäische Wirtschaftsforschung, Mannheim.
Bruck, A., 2011. Bull market for frugal engineering. High Growth Markets Magazine (October).
Carlsson, B., Stankiewicz, 1991. Nature, function and composition of technological systems. Journal of Evolutionary Economics 1 (2), 93–118.
Carlsson, B., Jacobsson, S., Holmén, M., Rickne, A., 2002. Innovation systems: analytical and methodological issues. Research Policy 31, 233–245.
Climate Group, 2011. Delivering Low Carbon Growth: A Guide to China's 12th Fiver Year Plan.
Coenen, L., Benneworth, P., Truffer, B., 2012. Toward a spatial perspective on sustainability transitions. Research Policy 41 (6), 968–979.
Dosi, G., Soete, L., 1988. Technical change and international trade. In: Technical Change and Economic Theory. Pinter, New York, pp. 401–431.
Dosi, G., Pavitt, K., Soete, L., 1990. The Economics of Technical Change and International Trade. Harvester Wheatsheaf, New York, http://www.lem.sssup.it/books.html
Elzen, B., Geels, F., Green, K., 2004. System Innovation and the Transition to Sustainability: Theory, Evidence and Policy. Edward Elgar, Cheltenham, UK.
Ernst, Young, 2012. Renewable energy country attractiveness indices: Issue 34.
European Commission, 2006a. Creating an Innovative Europe: Report of the Independent Expert Group on R&D and Innovation appointed following the Hampton Court Summit and chaired by Mr. Esko Aho. European Communities, Brussels.
European Commission, 2006b. European competitiveness report 2006. European Communities (COM(2006) 697 final), Brussels.
European Commission, 2006c. Putting knowledge into practice: A broad-based innovation strategy for the EU. European Commission (COM(2006) 502 final), Brussels.
European Commission, 2007a. 1–44 Explanatory Paper on the European Lead Market Approach: Methodology and Rationale (Annex II to COM(2007) 860 final).
European Commission, 2007b. Mid-term review of industrial policy: A contribution to the EU's Growth and Jobs Strategy. European Commission (COM(2007) 374 final), Brussels.
European Commission, 2010. Europe 2020: A strategy for smart, sustainable and inclusive growth, COM(2010) 2020 final.
European Commission, 2011. Innovation for a sustainable Future: The Eco-innovation Action Plan (Eco-AP), COM(2011) 899 final. Brussels.
Fagerberg, J., 1995. User-producer interaction, learning and comparative advantage. Cambridge Journal of Economics 19 (1), 243–256.
Fagerberg, J., 1996. Technology and competitiveness. Oxford Review of Economic Policy 12 (3), 39–51.
Federal Ministry of Education and Research, 2006. Die Hightech-Strategie für Deutschland. Federal Ministry of Education and Research, Berlin.
Federal Ministry of Education and Research, 2011. Wettbewerbsfähiger durch Leitmarktstrategie? Dokumentation des BMBF-Workshop vom 7. April 2011.
Federal Ministry of the Environment, 2006. Ökologische Industriepolitik: Memorandum für einen "New Deal" von Wirtschaft, Umwelt und Beschäftigung. Federal Ministry of the Environment, Berlin.
Geels, F.W., 2002. Technological transitions as evolutionary reconfiguration processes: a multi-level perspective and a case-study. Research Policy 31 (8/9), 1257–1274.
Geels, F., 2011. The role of cities in technological transitions: analytical clarifications and historical examples. In: Bulkeley, H., et al. (Eds.), Cities and Low Carbon Transitions. Routledge, London, pp. 13–28.
Government of Germany, 2008a. Entwurf eines Gesetzes zur Neuregelung des Rechts der Erneuerbaren Energien im Strombereich und zur Änderung damit zusammenhängender Vorschriften (Drucksache 16/8148). Deutscher Bundestag, Berlin.
Government of Germany, 2008b. Unterrichtung durch die Bundesregierung Masterplan Umwelttechnologien (Drucksache 16/11322). Deutscher Bundestag, Berlin.
Grin, J., Rotmans, J., Schot, J., 2010. Transitions to Sustainable Development: New Directions in the Study of Long term Structural Change. Routledge, London.
Hekkert, M.P., Suurs, R.a.a., Negro, S.O., Kuhlmann, S., et al., 2007. Functions of innovation systems: a new approach for analysing technological change. Technological Forecasting and Social Change 74 (4), 413–432.
Jacob, K., et al., 2005. Lead Markets for Environmental Innovations. Physika-Verlag, Heidelberg.
Jänicke, M., 2012. Dynamic governance of clean-energy markets: how technical innovation could accelerate climate policies. Journal of Cleaner Production 22 (1), 50–59.

Jänicke, M., Jacob, K., 2004. Lead markets for environmental innovations: a new role for the nation state. Global Environmental Politics 4 (1), 29–47.
Jänicke, M., Lindemann, S., 2010. Governing environmental innovations. Environmental Politics 19 (1), 127–141.
Kao, H.-h., Lee, J.-f., 2010. The application of shanzhai innovation model in China: the examples of mobile phone, notebook computer and automobile. In: Opening Up Innovation: Strategy, Organization and Technology, Imperial College London Business School, June 16–18, 2010.
Köhler, J., et al., 2013. Leaving fossil fuels behind? An innovation system analysis of low carbon cars. Journal of Cleaner Production 48, 176–186.
Kotabe, M., Helsen, K., 1998. Global Marketing Management. Wiley, New York.
Lundvall, B.-A. (Ed.), 1992. National Systems of Innovation: Towards a Theory of Innovation and Interactive Learning. Pinter, London.
Lundvall, B.-A., Johnson, B., 1994. The learning economy. Journal of Industry Studies 1 (2), 23–42.
Madsen, J.B., 2008. Innovations and manufacturing export performance in the OECD countries. Oxford Economic Papers 60 (1), 143–167.
Malerba, F., 2004. Sectoral Systems of Innovation. Concepts, Issues and Analysis of Six Major Sectors in Europe. Cambridge University Press, Cambridge, UK.
Malerba, F., 2006. Sectoral systems. How and why innovation differs across sectors. In: Fagerberg, J., Mowery, D., Nelson, R. (Eds.), The Oxford Handbook of Innovation. Oxford University Press, Oxford, pp. 380–406.
Mansfield, E., 1968. Industrial Research and Technological Innovation. Norton, New York.
Meyer-Krahmer, F., 1997. Innovation und Nachhaltigkeit. Ökologisches Wirtschaften (1), 20–23.
Meyer-Krahmer, F., 2004. Vorreiter-Märkte und Innovation: Ein neuer Ansatz der Technologie- und Innovationspolitik. In: Steinmeier, F.-W., Machnig, M. (Eds.), Made in Germany'21. Hoffmann und Campe, Hamburg, pp. 95–110.
Meyer-Krahmer, F., Reger, G., 1999. New perspectives on the innovation strategies of multinational enterprises: lessons for technology policy in Europe. Research Policy 28 (7), 751–776.
Murmann, J.P., Frenken, K., 2006. Toward a systematic framework for research on dominant designs, technological innovations, and industrial change. Research Policy 35 (7), 925–952.
Nationale Plattform Elektromobilität, 2011. Zweiter Bericht der Nationalen Plattform Elektromobilität. Berlin.
Nelson, R. (Ed.), 1993. National Innovation Systems. A Comparative Analysis, Oxford. Oxford University Press, Oxford.
Nelson, R., 1995. Co-evolution of industry structure, technology and supporting institutions, and the making of comparative advantage. International Journal of the Economics of Business 2 (2), 171–184.
Pietrobelli, C., Rabellotti, R., 2011. Global value chains meet innovation systems: are there learning opportunities for developing countries? World Development 39 (7), 1261–1269.
Porter, M., 1986. Changing patterns of international competition. California Management Review XXVIII (2), 9–40.
Porter, M., 1990. The Competitive Advantage of Nations. Free Press, New York.
Porter, M.E., Van Der Linde, C., 1995. Green and competitive: ending the stalemate green and competitive. Harvard Business Review (September–October), 120–134.
Quitzow, R., 2011. Integration von Nachhaltigkeitsaspekten in drei Programmen der EU-Innovationspolitik. In: Nachhaltigkeitsstrategien in Politik und Wirtschaft. Oekom, München, pp. 89–122.
Raffée, H., Kreutzer, R., 1989. Organisational dimensions of global marketing. European Journal of Marketing 23 (5), 43–57.
Rennings, K., Smidt, W., 2010. Rennings Smidt 2006 clean coal economia politica.pdf. Economia Politica XXVII (2), 303–328.
Rip, A., Kemp, R., 1998. Technological change. In: Human Choice and Climate Change. Batelle Press, Washington, DC, pp. 327–400.
Rogge, K.S., Hoffmann, V.H., 2010. The impact of the EU ETS on the sectoral innovation system for power generation technologies: findings for Germany. Energy Policy 38 (12), 7639–7652.
Rugman, A., 1991. Diamond in the rough. Business Quarterly 55 (3), 61.
Sanyal, P., 2004. The role of innovation and opportunity in bilateral OECD trade performance. Review of World Economics 140 (4), 634–664.
Sehgal, V., Dehoff, K., Panneer, G., 2010. The importance of frugal engineering. Strategy+Business (59).
Teece, D.J., 1986. Profiting from technological innovation: Implications for integration, collaboration, licensing and public policy. Research Policy 15 (February), 285–305.
Tiwari, R., Herstatt, C., 2012a. A Lead Market for Frugal Innovations? Extending the Lead Market Theory to Emerging Economies.
Tiwari, R., Herstatt, C., 2012b. Frugal Innovations for the 'Unserved Customer': An Assessment of India's Attractiveness as a Lead Market for Cost-Effective Products.
UNEP, 2012. Global Trends in Renewable Energy Investment 2012. Frankfurt School UNEP Coll*aborating Centre for Climate & Sustainable Energy Finance, Frankfurt.
Utterback, J., 1994. Mastering the Dynamics of Innovation. Harvard Business School Press, Boston.
Utterback, J.M., Abernathy, W.J., 1975. A dynamic model of process and product innovation. Omega 3 (6), 639–656.
Vernon, R., 1966. International investment and international trade in the product cycle. Quarterly Review of Economics 88, 190–207.
Vernon, R., 1979. The product cycle hypothesis in a new international environment. Oxford Bulletin of Economics and Statistics 54 (4), 255–268.
Von Hippel, E., 1986. Lead users: a source of nover product concepts. Management Science 32 (7), 791–806.
Von Hippel, E., 1994. The Sources of Innovation. Oxford University Press, Oxford.
Wakelin, K., 1998. Innovation and export behaviour at the firm level. Research Policy 26 (7/8), 829–841.
Walz, R., 2006. Impact of strategies to increase RES in Europe on employment and competitiveness. Energy & Environment 17 (6), 951–975.
Walz, R., Schleich, J., 2009. The Economics of Climate Change Policies: Macroeconomic Effects Structural Adjustments and Technological Change. Physika-Verlag, Heidelberg.

Walz, R., Köhler, J., 2014. Using lead market factors to assess the potential for a sustainability transition. Environmental Innovation and Sustainable Transitions 10, 59–76.
Windrum, P., Birchenhall, C., 1998. Is product life cycle theory a special case? Dominant designs and the emergence of market niches through coevolutionary-learning. Structural Change and Economic Dynamics 9, 109–134.
Yip, G., 1992. Total Global Strategy: Managing for Worldwide Competitive Advantage. Prentice Hall, Englewood Cliffs.
Zheng, J., Mehndiratta, S., Guo, J.Y., Liu, Z., 2012. Strategic policies and demonstration program of electric vehicle in China. Transport Policy 19 (1), 17–25.
Zhu, S., Shi, Y., 2010. Shanzhai manufacturing – an alternative innovation phenomenon in China: its value chain and implications for Chinese science and technology policies. Journal of Science and Technology Policy in China 1 (1), 29–49.